Learning Nonlinear Flow Matching for Multimodal Remote Sensing Image Clustering

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***Abstract*—Limited by different imaging mechanisms of sensors, multimodal remote sensing images hence form various data manifolds in observation spaces. Leveraging the multimodal complementarity is critical to improving clustering. In this article, we propose learning nonlinear flow matching (LNFM) that converts distributions of multimodal images to fuse multimodal features for clustering. Specifically, the nonlinear flow matching adaptively learns the nonlinear interpolation paths of the multimodal images and associated velocities, which are parameterized by neural networks. The learned nonlinear flow is a causal probability flow formalized as an ordinary differential equation (ODE). In practice, we train the LNFM model in latent spaces of the multimodal data for dimension alignment, and extract fusion features from the distribution paths by ODE sampling for *K*-means clustering. Significantly, unlike linear interpolation of rectified flow, the adaptive nonlinear paths avoid the interpolation crossing, leading to stable training loss and high-quality generation. Furthermore, we theoretically prove that the LNFM loss is bounded by the linear flow matching loss. To the best of our knowledge, this research could be one of the first flow matching models for multimodal clustering. Experiments on three benchmark datasets show that the proposed LNFM outperforms the relevant state-of-the-art (SOTA) methods in terms of accuracy and reveal its clustering and generation mechanisms.**

***Index Terms*—Feature fusion, multimodal clustering, nonlinear flow matching, remote sensing images.**

I. Introduction

OWING to a large imaging area, remote sensing images taken from unmanned aerial vehicle, aircraft, and satellite platforms provide rich land-cover information for Earth observation, leading to extensive applications in geological exploration and environmental monitoring [1], [2]. For the same scene, different sensors by wavelengths record diverse surface characteristics as multimodal data. Among the optical data, hyperspectral (HS) and multispectral (MS) images with a wide range of spectral channels provide clear textures of ground materials [3], while light detection and ranging (LiDAR) images carry corresponding height features [4]. In microwave remote sensing, synthetic aperture radar (SAR) images describe the rough terrain information [5]. Contrary to SAR, optical imaging without strong penetration is sensitive to clouds and mists. The complementary physical quantities of the multimodal images compensate for the shortcomings of single modalities [6], [7], having been forming a promising tendency for understanding semantics of land covers [8], [9].

Over the past decade, deep multimodal classification in a supervised manner has gained remarkable advances on remote sensing images [10], [11], [12], [13]. These classification models boil down to three main parts: feature extraction for individual modalities, feature fusion, and classifier on fused features. Under the framework, convolutional neural networks (CNNs) are first employed to extract multimodal features by [14], [15]. Following the CNN feature extractor, Wu et al. [16] design cross-channel reconstruction to yield compact fusion representations. Based on attention mechanisms, Li et al. [17] introduce a multimodal-cross attention network to capture second-order correlation of hidden features of optical and SAR images. In vision transformers, Roy et al. [6] propose a multimodal fusion transformer network with a multihead cross-patch attention to obtain better modal generalization. Considering characteristics of different modalities, Wang et al. [18] and Gao et al. [19] design a spectral–spatial feature learning model and a spatial–spectral perception network, respectively, to capture spatial features of SAR and spectral features of HS/MS. Pan et al. [20] improve HS and LiDAR classification by introducing a multimodal complementary information bottleneck with variational optimization. By self-supervised learning, Xue et al. [21] devise a generative model on multiple complementary views to get high-level meaningful feature representations. Similarly, self-supervised feature learning model [22] is equipped with an asymmetric autoencoder (AE) structure learning multimodal high-level key information. However, these attractive methods rely on classifier that requires varying degrees of manually annotated samples, bringing inconvenience to the application.

Conversely, multimodal clustering belongs to unsupervised learning without requiring label information [23], [24], and is equivalent to multiview clustering (MVC) in the machine learning community. The classic MVC algorithm [25] extends *K*-means of single views to multiview version, laying the foundation for multimodal clustering. Based on this, various machine learning methods are employed to develop associated multiview counterparts, including subspace clustering [26], spectral clustering (SC) [27], and nonnegative matrix factorization [28]. Among them, multiview subspace clustering (MVSC) methods have achieved remarkable success on multiview data, owing to the interpretability and robustness. MVSC regards different views as associated subspace sets, and collaboratively optimizes self-expression matrices of all

Received 8 January 2026; revised 9 March 2026, 15 April 2026, and 12 May 2026; accepted 12 June 2026. Date of publication 17 June 2026; date of current version 23 June 2026. This work was supported by Sichuan Science and Technology Program under Grant 2026NSFSC1405.

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Digital Object Identifier 10.1109/TGRS.2026.3704242

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views with certain norm constraints such that each matrix captures a subspace set. Then, fusing these representations produces an affinity matrix for SC. The original SC algorithm requires inputs with square matrix, which hence prevents from processing large-scale data. To address this issue, Kang et al. [29] perform MVSC on representative anchors instead of all data samples, producing anchor graph that is much smaller in size than a full connection graph. According to multiview similarity, Sun et al. [30] introduce unified anchors to MVSC, further reducing computational consumption. Different from independent anchor selection and anchor graph learning in previous methods, joint anchor and anchor graph optimization is integrated in one MVSC model [31], [32]. Regarding high-dimensional data and its balance structure, fast MVSC [33] jointly learns anchors, anchor graphs, and labels on embedding of data. Cheng et al. [34] proposed a large-scale MVSC also learning unified anchor and anchor graph on in embedding space, where the anchor graph induces clustering assignment matrix. Moreover, Yu et al. [35] improve MVSC by learning unified bipartite graph with sparse and low-rank constraints instead of unified anchor graph.

Nonetheless, there are only a few works that have focused on multimodal clustering of remote sensing images. For instance, TMPCC [36] uses contrastive loss and self-training loss to enforce a multimodal transformer to learn a discriminative shared space, where converged prototypes provide clustering soft labels. This method needs a large number of iterations and is sensitive to patch size. To improve sampling quality for contrastive learning, SEC-LSRM [37] builds AEs on superpixels to extract short-range information and graph AEs on pixel patches to obtain long-range information of multimodal data. At the kernel trick, AMKSC [38] maps original samples and anchors into high-dimensional kernel spaces, in which building MVSC model to learn unified anchor graph. DGLAP [39] applies superpixel segmentation algorithm to multimodal data to anchor, and then learn dual consensus anchor graphs that generate clustering labels. Similarly, based on anchors extracted by superpixel segmentation, FPFC [40] jointly optimizes two fuzzy clustering methods that project samples to anchors and anchors to clusters so that label matrix is achieved by the product of projection matrices. MSSAGF [41] performs superpixel segmentation and nonlinear neighborhood recovery for every modalities to get different spatial-aware anchors being used for learning multiscale spectral–spatial anchor graphs. MMAGL [42] applies low-pass graph filtering to modality-specific attributed graphs and then solve a fused graph for clustering constrained by four multiobjective clustering subproblems. SAGC [43] leverages spectral unmixing to propose subpixel anchor graph clustering. AMVFC [44] and MCSM [45] directly learn a consensus clustering matrix instead of anchor graph. Although the above anchor-based methods have achieved significant performances, they suffer from anchor number being difficult to determine.

The aforementioned clustering methods derive a consensus cluster representation for multimodal data by integrating representations from individual modalities. This process is inherently a form of manifold fusion and as a result faces challenges in capturing the complex distribution information characteristic of multimodal data. Since the distribution is defined on the manifold, it thus contains more representations for clustering than the manifold structure alone. Fortunately, the diffusion models serve as a distribution transport approach, and have attracted a lot of attention in recent years, thanks to the powerful generation capability [46], [47]. They divide the generative process into multiple steps, preserving distribution and leading to a promising direction for feature extraction [48]. Baranchuk et al. [49] first attempt to extract latent features from UNet of the diffusion model and then perform classification on the features. Similar to this study, Chen et al. [50] devise a spectral–spatial denoising network to improve spectral features of HS, which is fed to transformer-based encoder for classification. Unlike the two-stage manners, Jiang et al. [51] take multimodal remote sensing images as inputs of diffusion model that followed by a classifier, being trained jointly as an efficient pipeline. In clustering fields, diffusion subspace clustering [52] exploits noise prediction to augment self-representation of subspaces on HS images. Recently, conditional dual diffusion model [53] is developed for optical and SAR image clustering. It provides a decoupling AE conditioned by SAR images to fit noises and original optical images, preserving manifold structure of data, and the learned multimodal features extracted from the AE are used for K-means clustering. Recently, image fusion based on diffusion models has also seen tremendous advancements. Sun et al. [54] develop dual latent-guided diffusion to fuse infrared and visible images. By incorporating masked image modeling into a diffusion model, Tang et al. [55] transform the unsupervised image fusion task into dual masked image reconstruction. Despite this, diffusion models can only convert simple Gaussian distribution to empirical distribution of observation samples such that the distribution paths being a stochastic differential equation (SDE) suffer from random noises [56]. Hence, the features of the paths may affect the clustering.

Different from diffusion models, flow matching builds a bridge between any two distributions as an ordinary differential equation (ODE), so it can push a modality to another one [57], and lends itself naturally to image fusion by transporting source modal images to the fused modality. Wang et al. [58] devise an efficient rectified flow (RF) with guidance to fuse visible images. Similarly, Zhu et al. [59] collect fusion priors from multiple pretrained fusion models to supervise the training of the flow matching model. Flow matching learns velocity fields, along with the movement, which is equal to image generation. The distribution paths learned by flow matching correspond to a probability flow ODE that pushes one type of data to another type, preserving marginal distribution. Motivated by this, we exploit probability flow to build the transition paths of multimodal images, where the samples at the paths are regarded as fusion features. The fusion process is distinct from the aforementioned flow-based fusion approaches that consider the fused image as the target distribution. Obviously, they degenerate to single modal images when at the endpoints of the paths. The vanilla linear flow matching, i.e., rectified flow, uses linear interpolation of data as distribution paths, leading to the path crossing problem. Because each point at the

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intersections corresponds to multiple velocities, training the velocity estimator of distribution paths yields unstable losses that make generation worse. To address this issue, we take into account learning nonlinear flow matching (LNFM) for improving multimodal image clustering. Specifically, unlike existing flow matching methods, which require predefined or learning interpolation paths, the proposed LNFM method jointly learns nonlinear paths and associated velocities. The adaptive nonlinear paths automatically avoid the path crossing, yielding stable training loss. The path predictor and velocity estimator are parameterized by two networks. At sampling stage, we extract path–velocity features from the velocity network for multimodal clustering. In addition, we provide theoretical proof that the RF loss is an upper bound of LNFM loss. In summary, the core contributions of this study are presented as follows.

1) A promising nonlinear flow matching is proposed for multimodal clustering by jointly learning distribution paths and associated velocities.
2) The upper bound, causal ODE, and generalization of the proposed nonlinear flow matching are theoretically presented for interpretability.
3) The nonlinear flow matching is performed on multimodal remote sensing images, revealing the clustering mechanism.

The rest of the contents of this article are organized as follows. A brief view of preliminary work is given in Section II. The proposed nonlinear flow matching is presented in Section III. The theoretical analysis of the nonlinear flow matching is provided in Section IV. Experiments and conclusion are shown in Sections V and VI, respectively.

II. Preliminaries

A. Rectified Flow

For pushing source distributions $p(x_0) = \pi_0$ forward to target $p(x_1) = \pi_1$, where x_0 and x_1 are observation samples, the RF learns causal generative ODE of $\mathrm{d}z_t/\mathrm{d}t = v_\theta(z_t, t)$ from noncausal $\mathrm{d}x_t/\mathrm{d}t = x_1 - x_0$ of x_t inference. x_t is sampled by linear interpolation between x_0 and x_1 as

$$x_t = (1-t)\,x_0 + t x_1 \tag{1}$$

and v_θ is parameterized by a neural network and is solved by minimizing

$$\int_0^1 \mathbb{E}_{x_0,x_1}\left[\|x_1 - x_0 - v_\theta(x_t, t)\|^2\right] \mathrm{d}t. \tag{2}$$

Following the v_θ directions, then we can sample z_1 iteratively by

$$z_1 = z_0 + \int_0^1 v_\theta(z_t, t)\,\mathrm{d}t \tag{3}$$

where $z_0 = x_0$. z_t has the same marginal distributions as x_t, i.e., $p(z_t) = p(x_t)$.

Note that the loss of (2) is difficult to fit $v_\theta(x_t, t)$ if there are many x_t at intersections. This is because the intersection points x_t at t correspond to different velocity vectors $x_1 - x_0$, but $v_\theta(x_t, t)$ generates only one result.

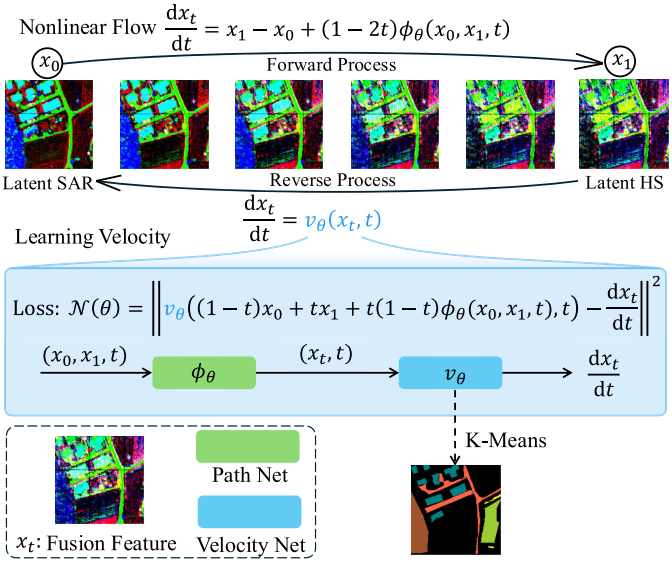


Fig. 1. Overview of the proposed LNFM method for multimodal clustering. Given multimodal images x_0 and x_1 in latent space, the LNFM simultaneously learns nonlinear interpolation paths x_t (forward ODE) and their velocity predictor v_θ (backward ODE). Sampling x_t by the learned ODE as multimodal fusion, the fusion features extracted from the velocity net are fed to K-means clustering.

B. Metric Flow Matching

Inspired by metric learning, metric flow matching (MFM) extends the RF interpolating along geodesics instead of straight lines. To this end, the nonlinear interpolants are defined as follows:

$$x_{t,\eta} = (1-t)\,x_0 + t x_1 + t(1-t)\,\varphi_\eta(x_0, x_1, t) \tag{4}$$

where φ_η is a parameterized neural network. Let $\mathcal{D} = \{x_i\}_{i=1}^N \in \mathbb{R}^d$ be a set formed by concatenating x_0 and x_1. $g(x) = G(x; \mathcal{D}) \in \mathrm{SPD}(d)$ is the Riemannian metric on d-dimensional manifold $(\mathcal{M}, g)$ parameterized by $\mathcal{D}$, where $G(x)$ is the Riemannian metric matrix. x_t fits the geodesics as finding their shortest distances in g on data manifold $\mathcal{M}$ by optimizing

$$\eta^* = \arg\min_\eta \mathbb{E}_{(x_0,x_1)\sim q,t}\left[\left(\frac{\mathrm{d}x_{t,\eta}}{\mathrm{d}t}\right)^\top G\left(x_{t,\eta}; \mathcal{D}\right)\left(\frac{\mathrm{d}x_{t,\eta}}{\mathrm{d}t}\right)\right] \tag{5}$$

such that x_{t,η^*} lies on $\mathcal{M}$ close to $\mathcal{D}$.

Once x_{t,η^*} is learned from (5), the velocity learning and distribution converting work like (2) and (3). Especially, since the geodesics of x_t lie on $\mathcal{M}$, the velocity estimator requires replacing norm $\|\cdot\|^2$ on ambient space with $\|\cdot\|_g^2$ on $\mathcal{M}$ as follows:

$$\mathbb{E}_{(x_0,x_1)\sim q,t}\left[\left\| v_\theta\left(x_{t,\eta^*}\right) - \frac{\mathrm{d}x_{t,\eta^*}}{\mathrm{d}t}\right\|^2_{g(x_t,\eta^*)}\right]. \tag{6}$$

However, this two-stage optimization makes the process of flow matching complex, and determining the suitable metric matrices G relies on the prior information of data.

III. Methodology

We consider fusing multimodal images using flow matching for clustering. To address the path crossing problem, LNFM

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Algorithm 1 Training

repeat

$t \sim \mathcal{U}(0,1)$

$\theta \leftarrow \theta - \eta\nabla_\theta \|x_1 - x_0 + (1-2t)\phi_\theta(x_0, x_1, t) - v_\theta((1-t)x_0 + tx_1 + (1-t)t\phi_\theta(x_0, x_1, t), t)\|^2$

until convergence

Algorithm 2 Sampling

for t **in** range$(0, 1, \Delta t)$:

$z_{t+1} \leftarrow z_t + \Delta t v_\theta(z_t)$

end for

with rigorous theoretical analysis is proposed. Then, we apply it to multimodal remote sensing image clustering. The overall framework of the proposed LNFM for clustering is illustrated in Fig. 1.

A. Problem Settings

Assume that x_0 and x_1 are aligned latent multimodal images. Linear interpolants of x_0 and x_0 usually incur intersections at x_t, which arises as an adversarial training problem of flow matching as follows:

$$\|x_1 - x_0 - v_\theta(x_t, t)\|^2. \tag{7}$$

The loss of (7) enforces v_θ to approximate the velocities of interpolation paths. However, the intersections of x_t correspond to $x_1 - x_0$ in a one-to-many way, but $v_\theta(x_t, t)$ just outputs one velocity vector. The path crossing increases training loss, resulting in unstable sampling effects in inference stage. Learning adaptive interpolation paths is key to avoiding the crossing, which affects the subsequent clustering task.

B. Learning Nonlinear Flow Matching

Considering latent multimodal images x_0 and $x_1 \in \mathbb{R}^{m\times n\times d}$, we are interested in pushing source distribution $p(x_0) = \pi_0$ forward to target distribution $p(x_1) = \pi_1$. In this way, x_t of the distribution paths can be regarded as fusion images of x_0 and x_1.

1) Interpolants: To avoid the interpolation crossing, we propose an adaptive nonlinear flow matching, the interpolants of which are given by

$$x_t = (1-t)\,x_0 + tx_1 + t\,(1-t)\,\phi_\theta\,(x_0, x_1, t)\,. \tag{8}$$

This equation corresponds to the ODE flow

$$\frac{\mathrm{d}x_t}{\mathrm{d}t} = x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_1, x_0, t) + t\,(1-t)\,\frac{\mathrm{d}\phi_\theta\,(x_0, x_1, t)}{\mathrm{d}t} \tag{9}$$

which is a noncausal system because of involving both x_0 and x_1. Note that, $\phi_\theta(x_0, x_1, t)$ of (9) has the same shape with x_0 and x_1 of $m \times n \times d$, while $\mathrm{d}\phi_\theta(x_0, x_1, t)/\mathrm{d}t$ follows t with the size of 1×1. Without explicit supervised information for training $\phi_\theta(x_0, x_1, t)$ with respective to t, the output of $\phi_\theta(x_0, x_1, t)$ is mainly determined by x_0 and x_1 rather than t because they lie in higher dimensional observation spaces, i.e., $m \times n \times d \gg 1 \times 1$. Hence, we consider $\phi_\theta(x_0, x_1, t)$ to be a constant with respect to t in practice. Since $\mathrm{d}\phi_\theta(x_0, x_1, t)/\mathrm{d}t \approx 0$, we simplify (9) as

$$\frac{\mathrm{d}x_t}{\mathrm{d}t} = x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_1, x_0, t) \tag{10}$$

which reduces the computational cost of the velocity field.

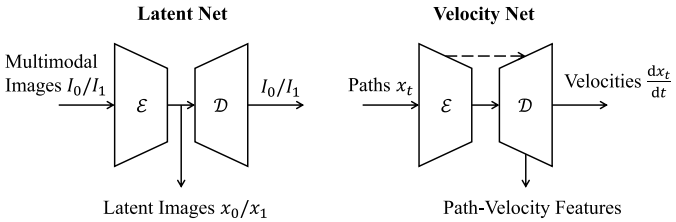


Fig. 2. Extracting latent images by the latent net (left) and path–velocity features by the velocity net (right).

2) Training: Different from the MFM method learning independent interpolants of (8) under manifold metric constraint, we jointly optimize the interpolation paths and their velocity estimator via learning a nonlinear flow matching

$$\begin{aligned}\mathcal{N}(x) &= \int_0^1 \mathbb{E}_{x_0,x_1}\left[\left\|\frac{\mathrm{d}x_t}{\mathrm{d}t} - v_\theta\,(x_t, t)\right\|^2\right]\mathrm{d}t \\ &= \int_0^1 \mathbb{E}_{x_0,x_1}\big[\,\|x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_0, x_1, t) \\ &\quad - v_\theta\,(x_t, t)\|^2\big]\,\mathrm{d}t\end{aligned} \tag{11}$$

where $v_\theta(x_t, t)$ serves as parameterized velocity estimator to fit the velocities (10). $\phi_\theta(x_0, x_1, t)$ recovers the nonlinear paths (8). From model fitting perspective, at intersections x_t, the nonlinear disturbance $\phi_\theta(x_0, x_1, t)$ of (11) is determined by adaptive learning, which alleviates the adversarial training of interpolation velocity $x_0 - x_1$ and velocity predictor $v_\theta(x_t, t)$, compared with (7).

3) Sampling: After that, converting the distribution from $p(x_0)$ follows a causal ODE without utilizing the future as follows:

$$\frac{\mathrm{d}z_t}{\mathrm{d}t} = v_\theta\,(z_t, t)\,, \quad z_0 = x_0 \sim \pi_0 \tag{12}$$

and sampling x_t can be achieved by applying (3). To sum up, the training and sampling process of the proposed LNFM method is shown in Algorithms 1 and 2 as pseudocode. Its properties and theoretical analysis will be discussed in detail in Section IV.

C. Learning Path–Velocity Features

1) Latent Space: Acknowledging that multimodal remote sensing images usually have different bands due to their imaging mechanisms, we are unable to build the interpolation paths between the original multimodal images with dimensional differences. To meet the dimension consistency required by the matching flow, we consider using AEs as a latent net to map them into latent spaces with unified dimensions. Let I be an original multimodal image. $\mathcal{E}$ and $\mathcal{D}$ denote the encoder and decoder of AEs, respectively. For each modality $i \in \{0, 1\}$,

$$\mathbb{E}_{I_i}\|I_i - \mathcal{D}_i \circ \mathcal{E}_i \circ I_i\|^2 \tag{13}$$

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Algorithm 3 LNFM for Multimodal Clustering

Input: I_0, I_1, t.
$x_0 \leftarrow AE(I_0), x_1 \leftarrow AE(I_1)$;
Applying **Algorithm** to x_0 and x_1;
Applying **Algorithm** 2 to generate z_t from $z_0 = x_0$;
Extracting path-velocity features of z_t and performing K-means to them.
Output: Clusters.

we extract latent images from the encoders of the trained latent nets by $x_i = \mathcal{E}_i \circ I_i$, which is illustrated in the left of Fig. 2. The encoder of the latent net is equipped with two-layer CNNs with dimensions of $[128, 4]$, and the decoder has a symmetric structure to the encoder. As a result, the proposed LNFM can perform on x_0 and $x_1 \in \mathbb{R}^{m\times n\times 4}$ embedded in the latent space.

2) Path and Velocity Nets: The interpolation paths x_t and their causal velocities $\mathrm{d}x_t/\mathrm{d}t$ involve the nonlinear disturbance $\phi_\theta(x_0, x_1, t)$ and the velocity estimator $v_\theta(x_t, t)$ as shown in (8) and (11). We parameterize them using the sought-after UNets as path net and velocity net, following the settings of the related work [53]. The dimension of the path net is set to $[16, 32, 64]$ for each encoder layer, while that of the velocity net is set to $[128, 64, 32, 16]$ for capturing intrinsic low-dimensional manifolds of the fusion features learned by LNFM. As we can see from the right of Fig. 2, the velocity net inputs paths x_t by encoder and outputs velocities $\mathrm{d}x_t/\mathrm{d}_t$ by decoder. Owing to the residual connections from the encoder to the decoder, the latter one captures both the path and velocity information of the multimodal images x_0 and x_1. Hence, we concatenate the outputs of four blocks of the decoder as path–velocity features required by clustering.

Note that the path features are stronger than the velocity features in terms of representation because of their many-to-one relationship. For example, without the crossing, at a time t, there are many points x_t on different interpolation paths, but they may be endowed with the same velocities learned by LNFM. Nevertheless, the path–velocity features integrate them and augment the single ones, leading to better discrimination for the downstream clustering task. Furthermore, any points of the linear interpolation paths have the same velocity through the derivative of a straight line, but the nonlinear paths induce various velocities at different points. Hence, LNFM enhances the path–velocity features of RF by improving the velocity diversity.

D. LNFM for Multimodal Clustering

It is well known that probability distributions are defined on manifolds; hence, they provide richer representation information than the manifold structure. The proposed LNFM adaptively learns the nonlinear distribution path of multimodal data such that the fused features extracted from this path capture not only the intrinsic manifold structures of multiple modalities but also the interpolation between their distributions, thereby improving cluster separability.

Remote sensing images usually have a large size containing rich spatial and band information but cause a heavy computational burden of LNFM at the same time. For this reason, the multimodal images are divided into several subimages of a size 64×64 without overlap. We perform the proposed LNFM method on the subimages as follows. First, we train the latent nets for each modality of the subimages by optimizing (13). Second, on the latent features x_0 and x_1 extracted from the latent nets, learning the path and velocity nets of the LNFM model yields the causal ODE of (12). Then, we use the ODE to generate the path–velocity features by applying (3). Specifically, given a t as the terminal time of the sampling paths, we sample x_t from x_0 through the ODE, and extract the path–velocity features from the velocity net corresponding to x_t, referring to Section III-C. Finally, feeding the path–velocity to the K-means algorithm produces the resulting clusters. The entire clustering process is provided in Algorithm 3.

IV. Theoretical Analysis

We present the preserving marginal distribution property and the probability ODE of the nonlinear flow matching, based on which its upper bound is provided. In addition, we analyze the relationship between the nonlinear flow matching and other probability models.

A. Marginal Distributions and Causal ODE

Given the differentiable random process x_t formed by nonlinear interpolation paths from x_0 to x_1

$$x_t = tx_1 + (1-t)\,x_0 + t\,(1-t)\,\phi_\theta\,(x_0, x_1, t) \tag{14}$$

the velocity field of (14) derives a probability flow as

$$\frac{\mathrm{d}z_t}{\mathrm{d}t} = v_\theta\,(z_t, t)\,, z_0 = x_0 \sim \pi_0 \tag{15}$$

where v_θ can be learned by minimizing the loss of (11). Thanks to [57, Th. 3.3], we know that x_t and z_t follow the same marginal distributions, i.e., $p(x_t) = p(z_t)$ such that the causal ODEs of (14) push z_t from distributions $p(x_0) \sim \pi_0$ to $p(x_1) \sim \pi_1$ without seeing the future.

To find the optimal v_θ of (15), we expand (11) as

$$\begin{aligned}
\mathcal{N}(x) &= \int_0^1 \mathbb{E}_{x_0,x_1}\big[\|x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_0, x_1, t) \\
&\qquad - v_\theta\,(x_t, t)\,\|^2\big]\,\mathrm{d}t \\
&= \int_0^1 \mathbb{E}_{x_t}\big\{\mathbb{E}_{x_0,x_1}\big[\|x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_0, x_1, t)\,\|^2 \\
&\qquad - 2\big[x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_0, x_1, t)\big]^\top \\
&\qquad \times v_\theta\,(x_t, t) + \|v_\theta\,(x_t, t)\,\|^2\big]\big\}\,\mathrm{d}t \\
&= \int_0^1 \mathbb{E}_{x_t}\big\{-2\mathbb{E}_{x_0,x_1}\big[x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_0, x_1, t) \\
&\qquad |x_t\big]^\top v_\theta\,(x_t, t) + \|v_\theta\,(x_t, t)\,\|^2\big\}\,\mathrm{d}t + C \\
&= \int_0^1 \mathbb{E}_{x_t}\big\{\|\mathbb{E}_{x_0,x_1}\big[x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_0, x_1, t)\,|x_t\big] \\
&\qquad - v_\theta\,(x_t, t)\,\|^2\big\}\,\mathrm{d}t + C'
\end{aligned} \tag{16}$$

which leads to the optimal solution

$$v_\theta^*\,(z_t, t) = \mathbb{E}_{x_0,x_1}\big[x_1 - x_0 + (1-2t)\,\phi_\theta\,(x_0, x_1, t)\,|x_t = z_t\big]. \tag{17}$$

Leveraging this formula, we are able to discuss the noncrossing property of the flow of (15) in Section IV-B.

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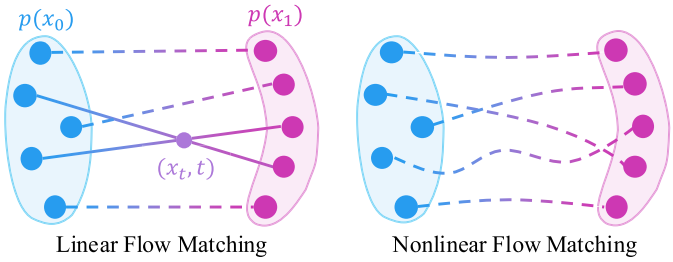


Fig. 3. Linear interpolation raises the crossing (left), while learning nonlinear paths leads to no intersections (right).

B. Upper Bound

We recall that the RF deduced from the linear interpolation $x_t = tx_1 + (1-t)x_0$ learns the velocity estimator by minimizing

$$\mathcal{L}(x) = \int_0^1 \mathbb{E}_{x_0,x_1}\left[\|x_1 - x_0 - v_\theta(x_t,t)\|^2\right] \mathrm{d}t \tag{18}$$

which derives its analytical solution $v_\theta^*(x_t,t) = \mathbb{E}_{x_0,x_1}(x_1 - x_0|x_t)$. Apparently, the nonlinear paths of (14) generalize the linear paths of the RF by adding a learnable item of ϕ_θ.

Using the following theorems, we then present an upper bound of the nonlinear flow matching loss of (11). Fig. 3 illustrates the main thoughts for the theoretical proof. The RF usually gives rise to the crossing interpolation paths that result in high training losses. To address this issue, we consider to learn nonlinear paths avoiding the crossing.

Definition 1 (Intersection Points): For any two interpolation paths x_t and $x'_{t'}, 0 \le t, t' \le 1$, we call that they intersect at x_t if $x_t = x'_{t'}$ and $t = t'$ hold.

Proposition 1 (Upper Bound of *(11)*): The nonlinear flow matching loss $\mathcal{N}(x)$ is bounded by the linear counterpart $\mathcal{L}(x)$ if $\phi_\theta(x_0,x_1,t)$ of $\mathcal{L}(x)$ is an injective function as follows:

$$\mathcal{N}(x) \le \mathcal{L}(x). \tag{19}$$

Proof: Let $x_t = \{x_t^+, x_t^-\}$ be all the linear interpolation points between x_0 and x_1, where x_t^+ denotes the intersection points as defined in Definition 1 and x_t^- is the other without the crossing. For any x_t^+, there exists a pair of (x'_0, x'_1) and (x''_0, x''_1) satisfying

$$(1-t)x'_0 + tx' = (1-t)x''_0 + tx'' = x_t^+ \tag{20}$$

but the injection ϕ_θ transforms them to yield $\phi_\theta(x'_0,x'_1,t) \neq \phi_\theta(x''_0,x''_1,t)$. Hence, we have

$$\begin{aligned}&(1-t)x'_0 + tx' + t(1-t)\phi\left(x'_0,x'_1,t\right)\\ &\quad\neq (1-t)x''_0 + tx'' + t(1-t)\phi\left(x''_0,x''_1,t\right).\end{aligned} \tag{21}$$

It indicates that through the nonlinear ϕ_θ, the nonlinear interpolations unfold the intersection points that the linear ones make, so we further obtain

$$\begin{aligned}&\mathbb{E}_{x_0,x_1}\left[x_1 - x_0 + (1-2t)\phi_\theta(x_0,x_1,t)\,|x_t^+\right]\\ &\quad= x_1 - x_0 + (1-2t)\phi_\theta(x_0,x_1,t).\end{aligned} \tag{22}$$

Since $\mathbb{E}_{x_0,x_1}(x_1 - x_0|x_t^-) = x_1 - x_0$, it yields $\|x_1 - x_0 - \mathbb{E}_{x_0,x_1}(x_1 - x_0|x_t^-)\|^2 = 0$. Inserting it and (22) into (18) and (11), we get

$$\begin{aligned}\mathcal{N}(x) &= \int_0^1 \mathbb{E}_{x_0,x_1}\left[\|x_1 - x_0 + (1-2t)\phi_\theta(x_0,x_1,t) - v_\theta^*(x_t,t)\|^2\right]\mathrm{d}t\\ &= \int_0^1 \mathbb{E}_{x_0,x_1}\{\|x_1 - x_0 + (1-2t)\phi_\theta(x_0,x_1,t)\\ &\qquad - \left[x_1 - x_0 + (1-2t)\phi_\theta(x_0,x_1,t)\right]\|^2\}\\ &= 0\\ &\le \int_0^1 \mathbb{E}_{x_0,x_1}\|x_1 - x_0 - \mathbb{E}_{x_0,x_1}\left(x_1 - x_0|x_t^+\right)\|^2 dt\\ &= \int_0^1 \mathbb{E}_{x_0,x_1}\|x_1 - x_0 - \mathbb{E}_{x_0,x_1}\left(x_1 - x_0|x_t^+\right)\|^2 dt\\ &\quad + \int_0^1 \mathbb{E}_{x_0,x_1}\|x_1 - x_0 - \mathbb{E}_{x_0,x_1}\left(x_1 - x_0|x_t^-\right)\|^2 dt\\ &= \int_0^1 \mathbb{E}_{x_0,x_1}\|x_1 - x_0 - v_\theta^*(x_t,t)\|^2 dt = \mathcal{L}(x).\end{aligned} \tag{23}$$

□

TABLE I
FORWARD PROBABILITY INSTANTIATIONS OF $p(x_t|x_0, x_T(x_0))$

	μ	σ_t^2
Brownian Bridge	$(1-\frac{t}{T})x_0 + \frac{t}{T}x_T$	$\frac{t(T-t)}{T}$
Stochastic Interpolants	$I(t,x_0,x_1)$	$\gamma^2(t)$
Rectified Flow	$(1-t)x_0 + x_1$	0
Nonlinear Flow	$(1-t)x_0 + x_1$	$[t(1-t)]^2$

Remark 1: The nonlinear ϕ_θ maps the endpoints of x_0 and x_1 into high-dimensional space, avoiding the path crossing, which is the key to the success of Proposition 1. $\mathcal{N}(x) \le \mathcal{L}(x)$ demonstrates that the nonlinear flow matching has better convergence and sampling than the linear version to learn the velocity estimator, verified in Section IV.

Remark 2: Apparently, minimizing $\mathcal{L}(x)$ of (11) with respective to θ forces $\phi_\theta(x_0,x_1,t)$ to be an injection that makes Proposition 1 hold, but in practice, it is difficult to achieve the optimal optimization owing to model design and training strategy of ϕ_θ.

C. Connection to Diffusion Bridges and Stochastic Interpolants

Unlike the denoising diffusion models [53] that convert sample distributions to observation distributions $p(x_0)$, the RF, diffusion bridge, e.g., Brownian bridge (BB) [60] and stochastic interpolant (SI) models [61] establish a random process between two different types of endpoints x_0 and x_1. They are therefore allowed to push any distributions to other ones.

Here, we discuss the relationship between the nonlinear flow matching and the other rich probability models. Given endpoints x_0 and x_1, their interpolation paths are given by SI $x_t = I(x_0,x_1,t) + z\gamma(t), z \sim \mathcal{N}(0,I)$, RF $x_t = (1-t)x_0 + x_1$, and NF $x_t = (1-t)x_0 + x_1 + t(1-t)\phi_\theta(x_0,x_1,t)$. We rewrite the forward probabilities $p(x_t|(x_0,x_1))$ in Gaussian conditional distribution paths [62], as shown in Table I, where let $\phi_\theta(x_0,x_1,t) \sim \mathcal{N}(0,I)$. Comparing the conditional probability models, when $T=1$ for BB and $I(t,x_0,x_1) = (1-t)x_0 + x_1$ for SI, they have the same means of linear interpolation, but

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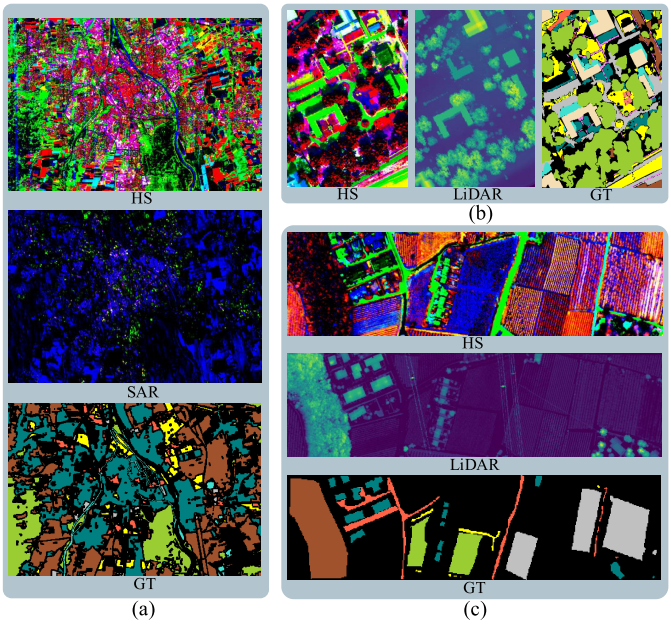


Fig. 4. Multimodal images in false colors and their GT labels of three datasets. (a) Augsburg. (b) MUUFL. (c) Trento.

are equipped with slightly different variances. The NF adds the randomness to the RF by the nonlinear disturbance ϕ_θ, and we can also convert NF into BB and SI forms through corresponding equation transformations.

D. Complexity

From Algorithms 1 and 2, the training and sampling processes require the inference of the path and velocity nets. They adopt UNet equipped with transformer mechanism that incurs the highest time complexity of $\mathcal{O}((m \times n)^2 d)$ for a image patch size of $m \times n$ and the largest feature dimension d of the transformer layers. Let T' and T'' denote the training epoch and the sampling step, respectively. Hence, the main time complexity of the proposed algorithms is given by $\mathcal{O}((T' + T'')(m \times n)^2 d)$. With the fast sampling techniques [63], [64], T'' can be reduced to a very small step size or even 1. Besides, the space complexity of the model depends on the configuration of UNet referring to Section III-C, and can be controlled within a reasonable range by the patch size for processing large-scale remote sensing images.

V. Experiments

We design a large number of experiments for evaluating clustering performance of the proposed LNFM model, and investigate its properties through parameter analysis, ablation study, and feature visualization.

A. Benchmarks

1) Datasets: Three benchmark datasets are adopted for the experiments, including Augsburg, MUUFL, and Trento. Their multimodal images, aligned according to land-covers, have the same spatial size and resolution, and are illustrated in Fig. 4 with associated false color images and ground truth (GT) maps.

TABLE II
Information of the Multimodal Datasets

Property	Augsburg		MUUFL		Trento	
	HS	SAR	HS	LiDAR	HS	LiDAR
Bands	180	4	64	2	63	2
Sizes	332 × 485		325 × 220		166×600	
Classes	7		11		6	

The Augsburg dataset consists of an HS image and a dual-Pol SAR image, captured by HySpex sensor and DLR3-3K system, respectively, over an urban area of Augsburg, Germany. The HS image is comprised of 180 bands covering a wavelength range of 0.4–2.5 μm, while the SAR image contains four features of VV and VH intensities, and real and imaginary parts derived from PolSAR covariance matrix. The MUUFL dataset contains an HS image with 64 bands of spectral resolution 10 nm in the spectral range 0.375–1.05 μm and the LiDAR image with two channels. They are acquired by CASI-1500 sensor with spatial resolution 0.54 × 1.0 m and Gemini ALTM LiDAR sensor with spatial resolution 0.6 × 0.78 m, over the University of Southern Mississippi, Long Beach, USA. The Trento dataset are comprised of an HS image with 63 bands in the wavelength range 0.4–0.98 μm, and the LiDAR image with two features. They are captured by AISA Eagle sensor and ALTM 3100EA sensor, over a rural region of the south of Trento, Italy, with spatial resolution 1 m. Overall, the band, size, and land-cover category information of the multimodal datasets is summarized in Table II.

2) Metrics: In order to measure the clustering performance, five popular metrics from various aspects are used in the experiments as follows: accuracy (ACC), Kappa coefficient (Kappa), normalized mutual information (NMI) [65], adjusted rand index (ARI), and purity (Purity), where the ACC, NMI, and Purity scores range in $[0, 1]$, while the Kappa and ARI scores lie in $[-1, 1]$. The metrics are calculated on the optimal matching of the predicted labels and the GT ones, producing better clustering performance with higher metric values. Besides, the time consumption of clustering methods is recorded in seconds (s) for evaluating the computational efficiency.

B. Baselines and Configurations

We select 13 typical or state-of-the-art (SOTA) multimodal clustering methods related to the proposed LNFM method for comparison experiments, highlighted as follows.

1) BMVC [66] is a binary MVC method by learning collaborative discrete representation and binary clustering structure.
2) SMVSC [30] is a scalable MVSC method based on unified anchors.
3) LMVSC [29] is a large-scale MVSC method using anchor graph instead of full graph.
4) MDC [67] is a multisensor clustering method that learns latent features for clustering using AEs.

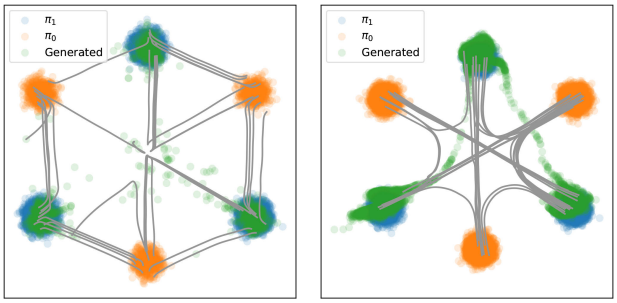
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5) FPMVS [32] jointly optimizes anchor selection and anchor graph for multiview subspace clustering.
6) FMVACC [68] is an MVC approach by anchor matching.
7) MSGL [69] is a scalable subspace clustering through learning a structured graph.
8) SDMVC [70] is a deep MVC method learning self-supervised features.
9) TMPCC [36] is a contrastive-learning-based multimodal clustering method.
10) GCFagg [71] improves multiview clustering by aggregating cross-sample and cross-view features.
11) FPFC [40] builds a dual multimodal fuzzy clustering method according to sample-anchor and anchor-cluster.
12) AMKSC [38] is a multimodal subspace clustering method with unified anchor learning and kernel trick.
13) CDD [53] is a diffusion-based multimodal clustering method preserving data manifolds.

For a fair comparison, we perform all comparison methods on pixel-level features. As features with a larger patch size carry more local spatial information, which tends to yield better clustering performance, the patch size of the TMPCC and the AMKSC methods is set to a minimum feasible value of 3 to reduce the impact of patches. The clustering results of the SDMVC and the GCFAgg methods are reported according to [37]. We set random seed to 42 for all methods. The other parameters of the comparison methods are determined according to the corresponding literature, or when no relevant data are available, they are tuned via grid search to achieve the best performance. As for the proposed LNFM method, the steps of its distribution paths are manually set to 50 at the training stage and 30 at the sampling (ODE solver) stage for computational efficiency. We set (x_0, x_1) to (HS, LiDAR), (LiDAR, HS), and (HS, SAR) for MUUFL, Trento, and Augsburg datasets, respectively. The model is trained on MUUFL, Trento, and Augsburg with epochs of 100, 200, and 250, respectively, the Adam optimizer with a learning rate of 0.0001, and batch sizes of 4 based on the image sizes. At the sampling stage, the terminal time step t is set to 11, 11, and 22 on MUUFL, Trento, and Augsburg, respectively, by applying grid search. All experiments are conducted on a Ubuntu server equipped with an Xeon Gold 5218 CPU, an RTX 2080Ti GPU, and 128 GB of RAM.

C. Toy Example

Our first goal is to explore how LNFM transmits distributions and avoids path crossings. To begin with, we adopt a 2-D mixture normal distribution dataset constructed by [57]. In Fig. 5, points $x_0 \sim \pi_0$ and $x_1 \sim \pi_1$ are used for training flow matching models that push from π_0 to π_1. LNFM, as an adaptive linear flow, is compared with the linear flow, known as the rectified flow [57]. The velocity net and the path net are equipped with a three- and two-layer multilayer perceptron (MLP), respectively, with hidden dimension of 100. To generate samples close to π_1 from π_0, we use the linear flow ODE and the nonlinear flow ODE to sample from x_0, and the sampling trajectories of 30 are recorded in gray as distribution paths. It clearly shows that LNFM prefers taking curves instead of straight lines to avoid path intersections, which verifies Proposition 1. Note that what appear as "intersections" in the figure are not intersections in the strict sense defined by the ODE, referring to Definition 1. Besides, the linear flow usually produces polyline paths because of interpolation crossing [Fig. 5(a)], which requires more sampling steps than the curved scheme of the nonlinear flow.



(a) (b)

Fig. 5. Distribution paths of generating samples from π_0 to π_1. (a) Linear flow. (b) Nonlinear flow.

D. Evaluation Results

1) Quantitative Performance: The objective clustering results of all comparison methods are displayed in Table III, where the best scores and the second-best scores on each metric are highlighted in bold and underline, respectively. From the table, the proposed LNFM achieves the overall best clustering performance according to the number of its best and second-best scores in the last column. Especially, it improves the second-best methods by 9.89%, 4.38%, and 5.99% scores with 0.8558, 0.7293, and 0.5714 ACC, respectively, which show the clustering superiority of LNFM against other comparison methods.

Compared with CDD that has a similar clustering mechanism to LNFM, the latter obtains better clustering results. This is because CDD introduces noise to the distribution paths, unlike LNFM, which directly converts a modality to another one, and also shows that the path–velocity features learned by LNFM are very discriminative for multimodal clustering. The patch-based methods, like the kernel subspace clustering method AMKSC and the contrastive learning method TMPCC, fail to gain satisfactory clustering results without large patch features, failing to capture spatial information. Even the Kappa, NMI, and ARI scores of TMPCC go to zero because all cover lands are clustered into one class. The anchor-based fuzzy clustering method FPFC achieves competitive results with LNFM on the Augsburg dataset, but performs worse than the latter on the Trento and MUUFL datasets in terms of all metrics, which may be caused by the unstable anchor selection.

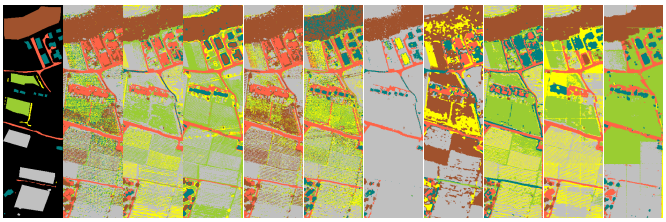
The traditional multiview methods, such as BMVC, FMVACC, and GCFagg, just reach below 72%, 58%, and 71% ACC of LNFM on the three datasets, respectively. The large gaps between them are attributed to the significant fusion features on the distribution paths of LNFM. The MVSC methods, including SMVSC, LMVSC, and FPMVS, are inferior to

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TABLE III

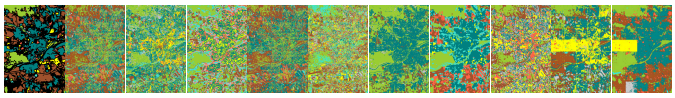
QUANTITATIVE EVALUATION OF 14 MULTIMODAL CLUSTERING METHODS ON THE TRENTO, AUGSBURG, AND MUUFL DATASETS IN TERMS OF THE OA, KAPPA, NMI, ARI, AND PURITY METRICS

Datasets	Metrics	Methods													
		BMVC	SMVSC	LMVSC	MDC	FPMVS	FMVACC	MSGL	SDMVC	TMPCC	GCFAgg	FPFC	AMKSC	CDD	LNFM
Trento	ACC	0.2293	0.6749	0.6566	0.7788	0.6944	0.6148	0.7629	0.6346	0.3476	0.5605	0.6418	0.7214	0.71	**0.8558**
	Kappa	0.0803	0.5769	0.5533	0.7014	0.5912	0.5232	0.6664	0.4834	0.	0.4225	0.5247	0.6541	0.6363	**0.8113**
	NMI	0.0264	0.5709	0.6221	0.6669	0.5375	0.5141	0.6859	0.3913	0.	0.4778	0.6397	0.7453	0.6798	**0.7507**
	ARI	0.0166	0.5819	0.5661	0.6379	0.5337	0.4231	0.6051	0.3822	0.	0.3292	0.5548	0.692	0.6563	**0.7328**
	Purity	0.3729	0.7377	0.7386	0.7863	0.7189	0.7201	0.792	0.6346	0.3476	0.5736	0.7176	**0.8724**	0.8477	0.8578
Augsburg	ACC	0.3571	0.3678	0.4771	0.5013	0.3961	0.2937	0.5823	0.4586	0.3874	0.4194	0.6987	0.3789	0.4791	**0.7293**
	Kappa	0.2357	0.1501	0.3237	0.3834	0.1708	0.1672	0.4176	0.1692	0.	0.2857	0.5994	0.2532	0.3184	**0.6218**
	NMI	0.2651	0.1514	0.2675	0.4307	0.1379	0.1438	0.3331	0.1629	0.	0.2389	**0.6081**	0.2609	0.3362	0.4935
	ARI	0.1702	0.0802	0.1985	0.3095	0.0847	0.0701	0.3014	0.2612	0.	0.1631	**0.5685**	0.165	0.2678	0.5292
	Purity	0.6468	0.5168	0.5919	0.7071	0.5057	0.5134	0.6233	0.4592	0.3874	0.5852	**0.8598**	0.5853	0.6528	0.8103
MUUFL	ACC	0.1655	0.379	0.4943	0.4391	0.3917	0.3671	0.5391	0.3977	0.433	0.4044	0.3928	0.4544	0.4913	**0.5714**
	Kappa	0.0795	0.2226	0.419	0.3643	0.23	0.2548	0.4624	0.3131	0.	0.3116	0.2797	0.3872	0.4131	**0.4762**
	NMI	0.0721	0.3611	0.5045	0.4518	0.3553	0.3213	**0.5262**	0.4422	0.	0.3303	0.2378	0.4855	0.4596	0.459
	ARI	0.0237	0.2228	0.3173	0.2237	0.2055	0.179	0.3535	0.2348	0.	0.1484	0.1532	0.2866	0.3155	**0.4409**
	Purity	0.4532	0.5782	0.7195	0.6713	0.5562	0.5571	**0.7499**	0.6564	0.433	0.601	0.5415	0.7131	0.7374	0.698



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 6. Cluster maps produced by different clustering models on the Trento dataset. (a) GT. (b) SMVSC. (c) LMVSC. (d) MDC. (e) FPMVS. (f) FMVACC. (g) MSGL. (h) FPFC. (i) AMKSC. (j) CDD. (k) LNFM.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 7. Cluster maps produced by different clustering models on the Augsburg dataset. (a) GT. (b) SMVSC. (c) LMVSC. (d) MDC. (e) FPMVS. (f) FMVACC. (g) MSGL. (h) FPFC. (i) AMKSC. (j) CDD. (k) LNFM.

LNFM, too. Although they efficiently fuse affinity matrices to achieve multimodal clustering, the graphs induced by the matrices lose high-dimensional information of data. Due to ignoring the correlation between multimodal data manifolds, the deep-learning-based methods, MDC and SDMVC, cannot provide remarkable clustering results. On the contrary, the LNFM method pushes the probability distributions of source modality to that of target modality to capture their intrinsic geometric structure.

2) Qualitative Performance: To investigate the clustering property of the proposed LNFM, we illustrate the cluster maps of ten comparison methods and GT labels in Figs. 6–8. The maps correspond to the clustering results shown in Table III. The unlabeled samples participate in clustering and hence are forcibly assigned annotated categories. As we



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 8. Cluster maps produced by different clustering models on the MUUFL dataset. (a) GT. (b) SMVSC. (c) LMVSC. (d) MDC. (e) FPMVS. (f) FMVACC. (g) MSGL. (h) FPFC. (i) AMKSC. (j) CDD. (k) LNFM.

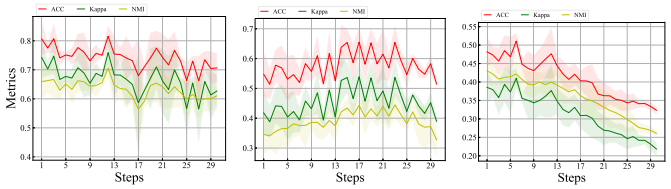
can see from the figures, the maps of LNFM are visually closest to their GT labels on three datasets, verifying the effectiveness of clustering on distribution paths. Compared with the recent SOTA methods of FPFC and AMKSC, LNFM leads to smoother clustering maps preserving the local spatial consistency of the land covers. This property of LNFM is also reflected in clustering of unlabeled areas, which makes sense for dealing with real scenarios. The dual conditional diffusion model CDD performs on the subimages as well as LNFM, but yields more distinct boundaries between the patch clustering maps, which conveys the great potential of LNFM for processing large images. Seeing from Fig. 7(k), two adjacent patch images were given almost the same color (yellow) because their most pixels are attracted by a few internal samples into the same cluster, indicating that the sample imbalance issue remains a challenge.

3) Time Consumption: The computational time consumption of 12 comparison methods on the three datasets is presented in Table IV. LNFM takes more time than most comparison methods and less time than the contrastive learning TMPCC. The biggest time consumption of LNFM is concentrated on training the nonlinear flow matching as discussed in Section IV-D. Fitting the velocities of the distribution paths between multimodal images needs a large number of iterations. This is different from CDD that learns the distribution maps from the Gaussian sampling to a modality conditioned

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TABLE IV
TIME CONSUMPTION OF DIFFERENT COMPARISON METHODS ON THREE MULTIMODAL DATASETS IN SECONDS (s)

Datasets	BMVC	SMVSC	LMVSC	MDC	FPMVS	FMVACC	MSGL	TMPCC	FPFC	AMKSC	CDD	LNFM
Trento	7.62	97.0	57.99	72.96	161.37	124.98	289.20	938.24	8.35	30.25	92.04	624.13
Augsburg	18.78	160.19	126.18	73.90	302.86	262.76	504.32	1910.93	16.83	54.45	142.48	1130.99
MUUFL	5.92	86.11	54.63	33.06	123.78	102.17	262.99	712.53	7.09	24.97	75.03	245.58



(a) (b) (c)

Fig. 9. Clustering performance of the proposed LNFM with different terminal time steps t in the distribution paths. (a) Trento (x_0: LiDAR, x_1: HS). (b) Augsburg (x_0: HS, x_1: SAR). (c) MUUFL (x_0: HS, x_1: LiDAR).

TABLE V
CLUSTERING PERFORMANCE OF THE RECTIFIED FLOW AND THE PROPOSED LNFM

Datasets	Methods	OA	Kappa	NMI	ARI	Purity
Trento	Rectified flow	0.7257	0.6347	0.5853	0.6243	0.7559
	LNFM	**0.8558**	**0.8113**	**0.7507**	**0.7328**	**0.8578**
Augsburg	Rectified flow	0.5855	0.4699	0.4115	0.3812	0.7716
	LNFM	**0.7293**	**0.6218**	**0.4935**	**0.5292**	**0.8103**
MUUFL	Rectified flow	0.4621	0.3698	0.4081	0.301	0.6976
	LNFM	**0.5714**	**0.4762**	**0.459**	**0.4409**	**0.698**

by another modality. Apparently, for improving clustering efficiency, shortening the interpolation paths of LNFM can significantly reduce training time, and increasing the sampling step length can lower the generation time for extracting the fusion features. From the perspective of ODE sampling, more sampling steps tend to yield better generated features and consequently enhance the clustering representations. Alternatively, learning uniform velocity of the distribution path instead of instantaneous velocity enables one-step sampling without sacrificing generation quality [72].

a) Distribution paths: The diffusion model of CDD and the LNFM model extract the fusion features from the distribution path $p(\pi) \to p(x_1)$ (conditioned by x_0) and $p(x_0) \to p(x_1)$, respectively, where $p(\pi)$ denotes the Gaussian distribution. $p(x_i), i = 0, 1$ stands for the data distribution of each modality. Due to the simple distribution $p(\pi)$, learning the diffusion path is much easier than the path between two complex distributions of $p(x_0)$ and $p(x_1)$, so the CDD consumes more training time and sampling time than the LNFM, referring to Table IV. However, the diffusion approach suffers from Gaussian noise, which more or less disrupts the manifold structure of the fused data. In contrast, LNFM directly maps from $p(x_0)$ to $p(x_1)$ without requiring additional guidance, resulting in significantly improved clustering performance compared to CDD, as shown in Table III, while also offering a more intuitive fusion process.

E. Effects of Terminal Time t for Sampling

The proposed LNFM involves an important parameter t, i.e., the terminal time for sampling, which determines the fusion features for clustering on the distribution paths, as described in Section III-D. Given a modality x_0, the LNFM model through ODE pushes empirical distribution from $p(x_0)$ to $p(x_1)$ along with $t \in [0, 1]$. To study the impact of the distribution paths, we visualize the clustering performance of LNFM in mean and standard deviation of three metrics at different terminal time steps as shown in Fig. 9, where the results were collected from ten repeated runs without fixing the random seed. Recall that x_0 is set to LiDAR on the Trento dataset, HS on the Augsburg dataset, and HS on the MUUFL dataset, while the remaining modalities of the three datasets are assigned to x_1. As we can see from the figure, the best scores of LNFM are on the distribution paths rather than at either end of the paths. This phenomenon verifies the correctness of the motivation behind LNFM. That is, the distribution paths between the multimodal images x_0 and x_1 can be seen as a multimodal feature fusion used for clustering. Moreover, the tendency of the metric curve reflects the importance of multimodal images. For example, on the Augsburg dataset, the higher clustering scores are closer to the side of SAR (x_1), while those trends to the side of LiDAR (x_0) and the side of HS (x_0) on Trento and MUUFL datasets, respectively. Affected by the training with random sampling t, the variance values reflect a certain degree of instability of the model, which could be effectively alleviated by exponential moving average (EMA) optimizer [57].

F. Ablation Study

1) Interpolation Paths: In Section III-B, we claim that owing to avoiding the interpolation crossing, the proposed nonlinear flow matching (LNFM) has a smaller loss than its linear counterpart (rectified flow). An inequality relationship between their losses is given by Proposition 1. To confirm this, we illustrate their training losses on the three datasets Fig. 10, where the linear version and the nonlinear version are drawn in blue and red, and the generated features x_1 at an iteration by their ODEs are displayed over the losses. Observing the training performance, it is easy to find that the losses yielded by LNFM are more stable than those produced by rectified flow, especially on the Augsburg dataset. Besides, the generated samples of LNFM are closer to the GT labels on the three datasets, referring to Fig. 4. These analyses have validated the stability of LNFM due to learning the nonlinear interpolation paths, and exactly confirm Proposition 1.

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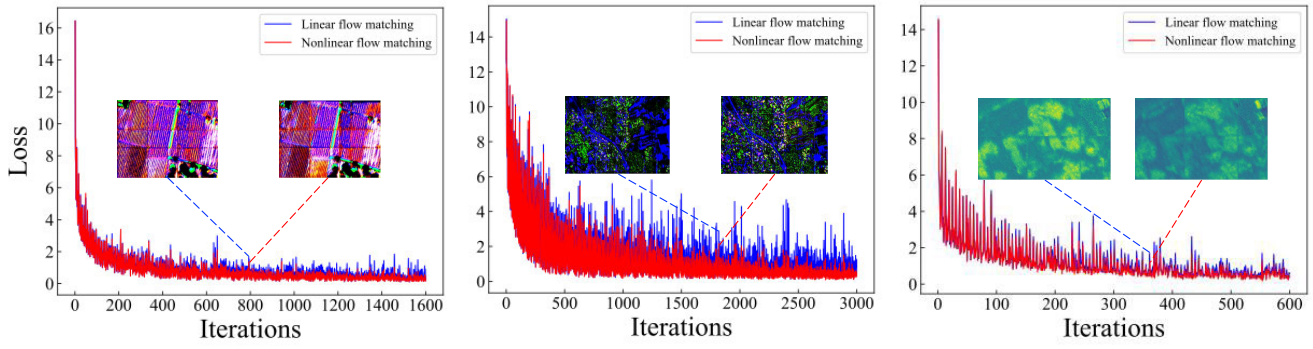


Fig. 10. Training stability of linear flow matching (rectified flow) versus that of nonlinear flow matching (LNFM). (a) Trento. (b) Augsburg. (c) MUUFL.

TABLE VI
CLUSTERING PERFORMANCE WITH DIFFERENT MODAL DATA

Datasets	Metrics	OA	Kappa	NMI	ARI	Purity
Trento	HS	0.4495	0.292	0.2694	0.2029	0.5822
	LiDAR	0.7699	0.6739	0.6407	0.5831	0.7711
	HS+LiDAR	**0.8558**	**0.8113**	**0.7507**	**0.7328**	**0.8578**
Augsburg	HS	0.4362	0.2626	0.2029	0.127	0.5494
	SAR	0.5477	0.3366	0.2946	0.2039	0.6018
	HS + SAR	**0.7293**	**0.6218**	**0.4935**	**0.5292**	**0.8103**
MUUFL	HS	0.3538	0.2501	0.2869	0.1824	0.5747
	LiDAR	0.3298	0.2412	0.2881	0.1423	0.5629
	HS+LiDAR	**0.5714**	**0.4762**	**0.459**	**0.4409**	**0.698**

Based on the training comparison, we further conduct an ablation study of the interpolation paths by clustering performance. Table V shows the clustering results of two flow matching methods with linear interpolation paths and nonlinear ones. From the table, LNFM significantly outperforms rectified flow by large OA improvements, and achieves the best scores in all metrics. This result is consistent with the comparison of the training losses. In summary, the ablation study demonstrates that learning nonlinear interpolation paths to avoid the crossing can improve training stability and clustering performance.

2) Modality: To verify the effectiveness of the fusion features, we provide ablation study of different modal features. Specifically, we perform K-means algorithm on the latent features generated by the latent nets for each single modal image, and on the fusion features extracted from the velocity net of LNFM for the multimodal image. Their clustering results are given in Table VI. The highlighted scores indicate the superiority of the fusion features over the single modal features. The HS features produce comparable results to the SAR features on MUUFL dataset, while leading to lower scores than other single modal features do on the Trento and Augsburg datasets because the high-dimensional manifolds of HS data increase feature redundancy.

The embeddings of the multimodal images and the fusion features on the Trento dataset are visualized in Fig. 11. To achieve the 2-D manifolds, we apply t-SNE algorithm to the

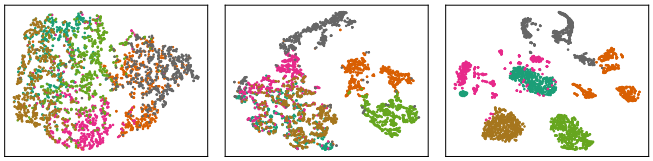


Fig. 11. Visualization of multimodal embeddings on the Trento dataset. (a) HS data. (b) LiDAR data. (c) Learned fusion features by the LNFM model.

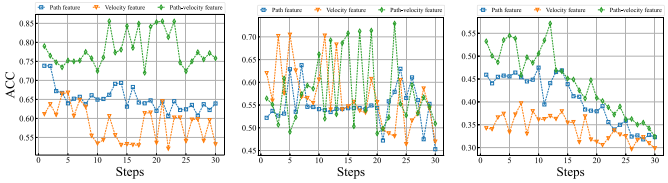


Fig. 12. Clustering performance with different features learned by the LNFM model on three datasets. (a) Trento. (b) Augsburg. (c) MUUFL.

latent features of single modal images and the path–velocity features of the multimodal images. Comparing Fig. 11(a) and (b), the LiDAR embeddings are more discriminative than the HS embeddings where the clusters overlap each other. This result leads to better clustering performance of the LiDAR features, which is already verified in Table VI. From Fig. 11(c), LNFM learns the fusion features with smaller intracluster distance and larger intercluster distance against other single modal features.

3) Path–Velocity Features: As described in Section III-C, the path–velocity features are extracted from the velocity net that accepts the samples of the interpolation paths, and outputs their associated velocity vectors. Hence, the velocity net contains both the interpolation path and velocity information between the multimodal images x_0 and x_1. We do an ablation study of the path–velocity features to understand their contribution to clustering. Specifically, following the training and sampling of LNFM for clustering on the three datasets, we extract the path features and the velocity features from the encoder and the decoder of the velocity net without the residual connections, respectively. Applying K-means to the

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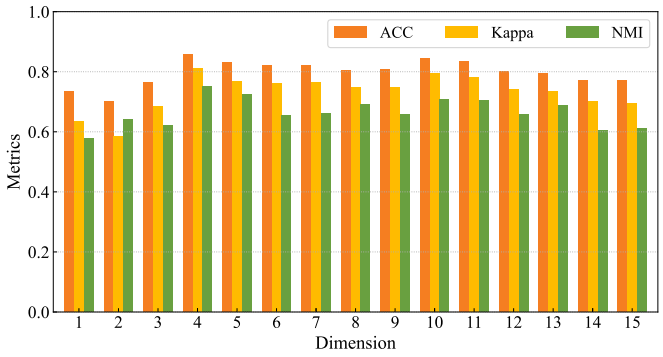


Fig. 13. Impact of the latent space dimensionality for clustering performance on Trento dataset.

three kinds of fusion features yields clustering performance as plotted in Fig. 12. On the Trento and MUUFL datasets, the clustering results for the path–velocity features are almost all higher than those for the path features and the velocity features. Therefore, we know that the path–velocity features have strengthened the single-type features of the interpolants. Similarly, the path features are superior to the velocity features on the two datasets, as the analysis mentioned in Section III-C that under the no-crossing situation, the paths and the velocities are one-to-many such that the path features are more discriminative than the velocity ones. Although the tree features do not make a significant difference in clustering performance, the path–velocity still achieves the best ACC at around time step 24 of the distribution path.

4) Latent Space Dimensionality: Section III-C has discussed how to embed multimodal images into the latent spaces of a uniform dimension. Here, we investigate the impact of dimensionality on multimodal clustering. To this end, Fig. 13 illustrates the clustering performance of the LNFM method on three metrics using the Trento dataset when varying the dimensions from 1 to 15. Overall, varying the dimension yields stable clustering performance. While the optimal dimension is concentrated around 4, extremely low or high dimensions (e.g., 1 or 15) result in inferior clustering results. This is because compressing the high-dimensional HS image into a lower dimensional latent space incurs information loss, and mapping the low-dimensional LiDAR data into this space leads to data sparsity. An extreme dimensionality of the latent space hinders the manifold alignment of multimodal data. Thus, we conjecture that an appropriate dimensionality falls between the respective channel counts of the multimodal data.

G. Feature Analysis

1) Analysis of Fusion Images by Sampling: On the latent features generated by the latent nets, we sample z_t from $z_0 = x_0$ to z_1 by performing the trained LNFM model, and illustrate the interpolation paths of z_t. Fig. 14 gives an example of the path visualization in false colors on the Trento dataset. Let x_0 be the HS image and x_1 be the LiDAR image. We apply Algorithm 2 to generate the fusion images z_t from to z_0 in the latent space. As we can see from the figure, three fusion images $z_{1/3}$, $z_{2/3}$, and z_1 sampled by the model are displayed at $t \in \{1/3, 2/3, 1\}$.

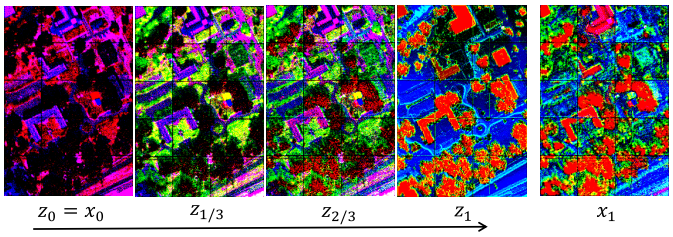


Fig. 14. Sampling z_t from $z_0 = x_0$ to z_1 through the ODE of the trained LNFM model, where the generated z_1 is close to the reference x_1 on the MUUFL dataset.

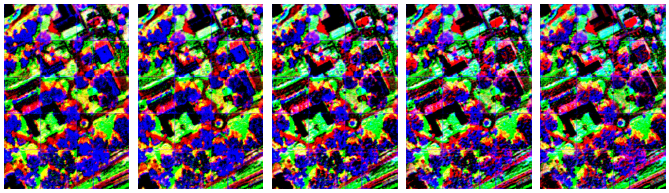


Fig. 15. Visualization of $\phi(x_0, x_1, t)$ features at different time steps on the MUUFL dataset. (a) $t = 1$. (b) $t = 8$. (c) $t = 15$. (d) $t = 22$. (e) $t = 29$.

The latent images x_0 and x_1 are very different because of the modal differences. As t increases, the generated z_t gradually approximates x_1. Consequently, z_1 is very close to x_1, and the semantics of $z_{1/3}$ and $z_{2/3}$ are between that of x_0 and x_1. This property of the sampling path meets the expectation of multimodal feature fusion for clustering, which has already been verified in Fig. 9(c). The experiment demonstrates that the proposed LNFM model not only generates high-quality images but also yields discriminative fusion features for clustering. Because the adaptive distribution paths avoid the crossing, the training loss of LNFM becomes more stable, which makes it a promising generative model.

2) Visualization of ϕ: The nonlinear disturbance ϕ of the linear interpolation of x_0 and x_1 is the key to achieving the nonlinear interpolation path learning, so exploring its effects on clustering is necessary. For this purpose, we illustrate the outputs of ϕ with the first three dimensions on the MUUFL dataset as illustrated in Fig. 15. From left to right, each subfigure is generated by $\phi(x_0, x_1, t)$ at different time steps $t \in \{1, 8, 15, 22, 29\}$ and the endpoints x_0 and x_1. Referring to Fig. 4, we find that the disturbance term captures basic shapes of the land covers such that the nonlinear interpolants $x_t = (1-t)x_0 + tx_1 + t(1-t)\phi_\theta(x_0, x_1, t)$ provide the additional features to their linear counterpart. It enhances the representation of the interpolation paths for clustering. Furthermore, the features of ϕ_θ have very slight differences along the paths at different times, which also verifies the assumption $\mathrm{d}\phi_\theta(x_0, x_1, t)/\mathrm{d}t \approx 0$ discussed in Section III-B. Since the coefficient $t(1-t)$ of ϕ_θ is continuous, $t(1-t)\phi_\theta(x_0, x_1, t)$ is approximately smooth, which allows LNFM to generate samples efficiently using a big step of the ODE sampler. It is no surprise that the nonlinear ϕ_θ by adaptive learning cannot capture the manifold structure of the multimodal data without any geodesic metrics. On the contrary, it yields a stable representation and leads to smooth interpolation paths for accelerating sampling.

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VI. Conclusion

To achieve discriminative features for multimodal remote sensing image clustering, we propose to learn nonlinear interpolation paths of the multimodal images as the fusion features. Based on this thought, an LNFM method is devised, avoiding the path crossing. Accordingly, we theoretically prove that the linear flow matching loss is an upper bound of the LNFM loss. Extensive comparison experiments demonstrated the superiority of LNFM over other SOTA multimodal clustering methods by the best ACC scores of 0.8558, 0.7293, and 0.5714 on Trento, Augsburg, and MUUFL datasets, respectively. Qualitative evaluation comparison shows that LNFM tends to produce smoother clustering maps, but suffers from the sample imbalance problem. The clustering performance of fusion features at different time steps verifies the effectiveness of the LNFM model again. Without path intersection, it leads to stable training loss and high-quality image generation. Moreover, the ablation study shows the effects of the nonlinear interpolation paths, modality, and path–velocity features for clustering. Through the feature analysis, we find that the proposed model can generate smooth distribution paths for quick sampling.

Besides, in virtue of being equivalent to distribution interpolation, the proposed method exhibits a potential resilience to such multimodal misalignment, noise, or missing modality problems. When multimodal data serve as a conditioning input to the flow matching model, the clustering framework, initially designed for two modalities, can be seamlessly scaled to a multimodality version. Unlike conditional diffusion models, training the generative distribution paths needs to take a lot of time with many iterations because it is difficult for flow matching to fit the conversion from one modal distribution to another. Hence, more efficient nonlinear flow matching will be preferentially considered in the future. Furthermore, the choice of the terminal time step t determines the weights and fusion effect of multimodal images, which is a challenging problem. Drawing on multiobjective optimization, treating t as an additional optimization variable to adaptively pursue the terminal time step is a very promising direction.

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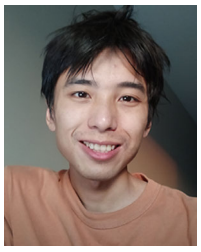
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